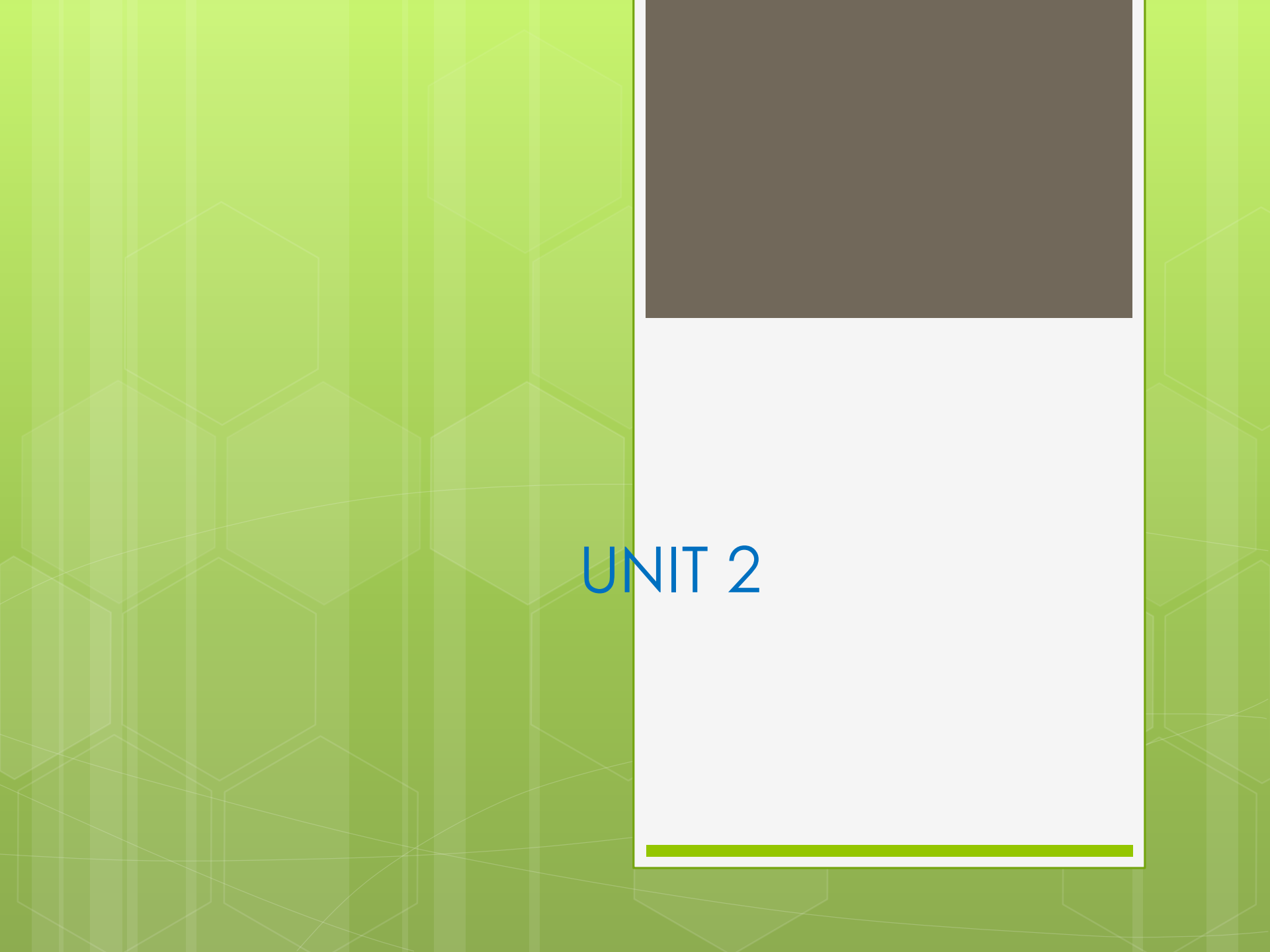




UNIT 2

Matter

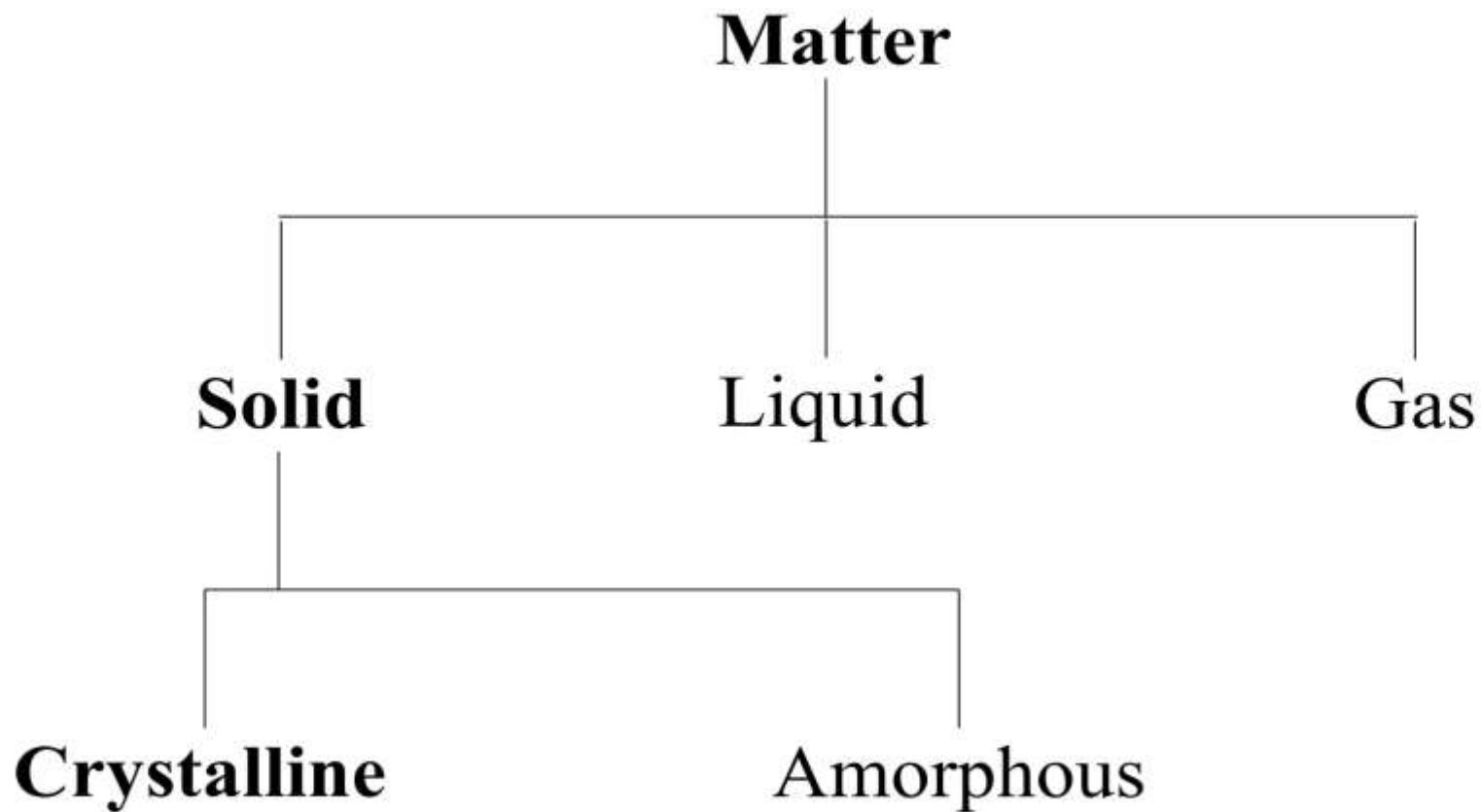
Solid

Liquid

Gas

Crystalline

Amorphous



Crystal=lattice+basis

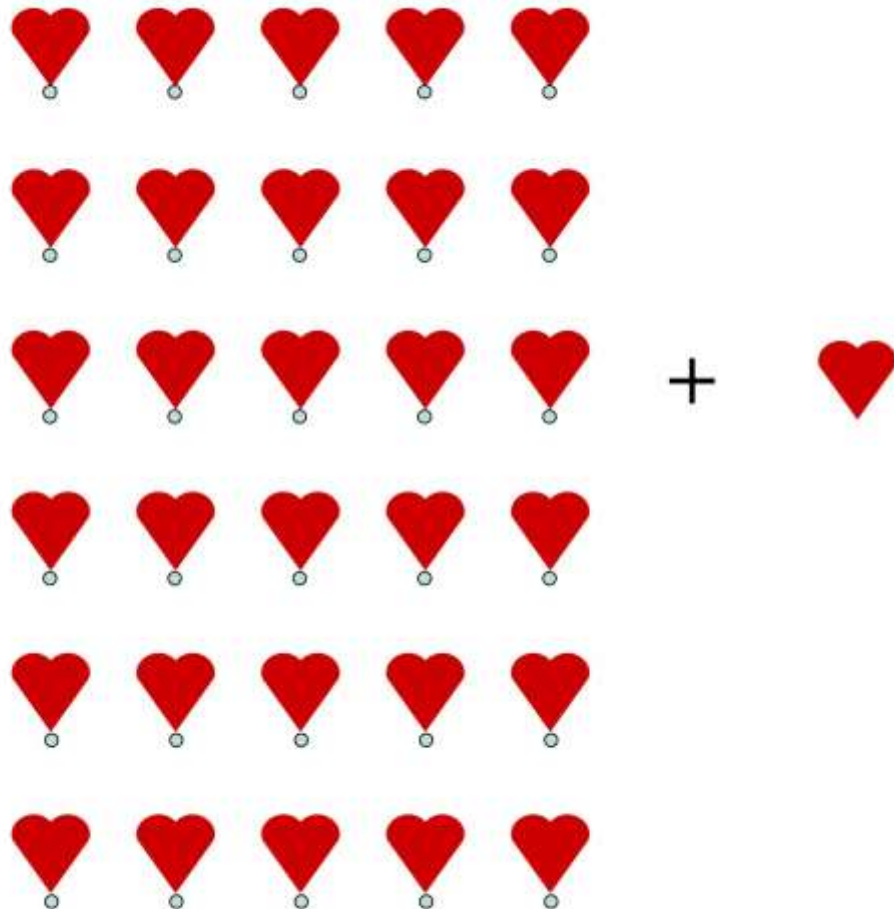
Lattice: the underlying periodicity of the crystal,

Basis: atom or group of atoms associated with each lattice points

Lattice: how to repeat

Motif: what to repeat

Love Pattern = Love Lattice + Heart



What is the relation between
the two?

$$\text{Crystal} = \text{Lattice} + \text{Motif}$$

Motif or basis: an atom or a
group of atoms associated
with each lattice point

Crystal

A 3D
translationally
periodic
arrangement
of **atoms**

Lattice

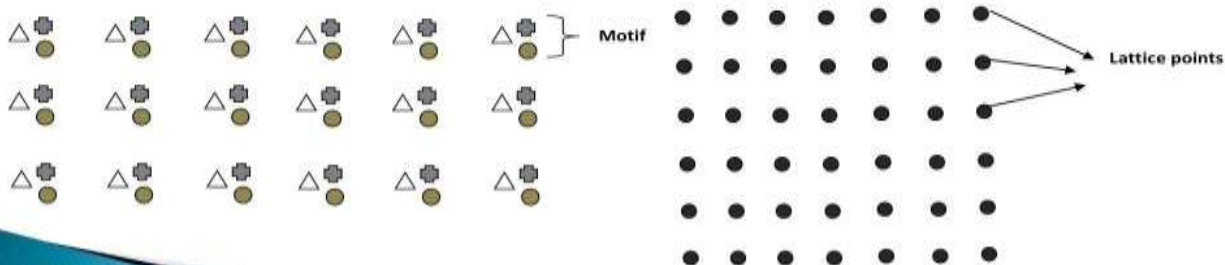
A 3D
translationally
periodic
arrangement of
points

Space Lattice

A *discrete* array of points in 3-d space such that every point has *identical surroundings*

SPACE LATTICE

- ▶ A crystal structure is built of infinite repetition of identical atoms or group of atoms in three dimensions.
- ▶ A group of atoms is called the basis or motif.
- ▶ The set of mathematical points to which the basis is attached is called a lattice.



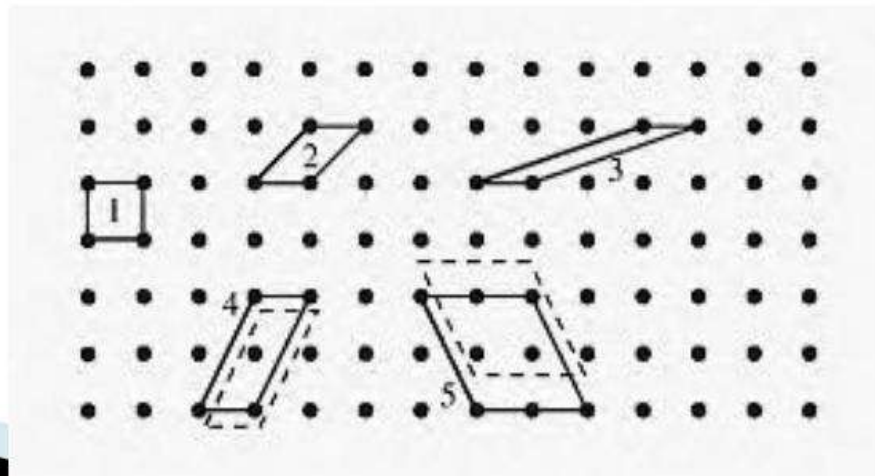
CRYSTAL STRUCTURE

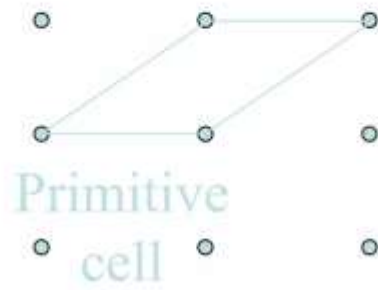
- **Crystal**— considered as consisting of tiny blocks which are repeated in 3D pattern.
- Tiny block--- **UNIT CELL**
Unit cell---arrangement of small group of atoms . It is that volume of solid from which entire crystal can be constructed by repeated translation in 3D.
- **Lattice point** --- each atom in cell is replaced by point.
- **Space lattice(infinite lattice array)** --- arrangement of lattice in 3D.every point has identical surrounding.
- **Lattice spacing**--- distance between atom points.



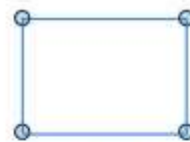
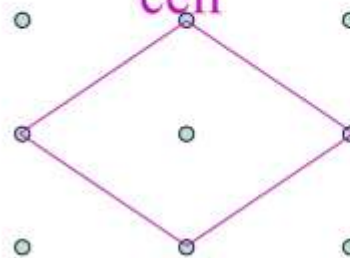
Unit cell

- ▶ The parallelepiped-shaped volume which, when reproduced by close packing in three dimensions, gives the whole crystal structure or space lattice itself is called the *unit cell*.
- ▶ Definition: *Unit cell is the smallest geometrical volume in a space lattice, defined by three basis vectors, which on translation along three basis vector directions yields the complete crystal structure or space lattice itself.*





Nonprimitive
cell



Primitive
cell

Cells

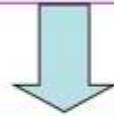
- ❖ A cell is a finite representation of the infinite lattice
- ❖ A cell is a parallelogram (2D) or a parallelepiped (3D) with lattice points at their corners.
- ❖ If the lattice points are only at the corners, the cell is primitive.
- ❖ If there are lattice points in the cell other than the corners, the cell is nonprimitive.

Lattice Parameters

Lengths of the three sides of the parallelopiped :
a, b and c.

The three angles between the sides: α , β , γ

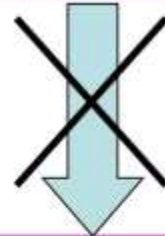
The six lattice parameters $a, b, c, \alpha, \beta, \gamma$



The cell of the lattice

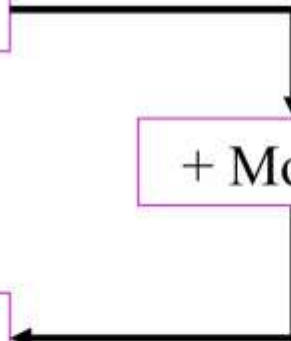


lattice



crystal

+ Motif

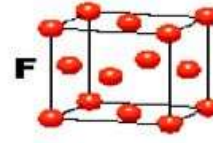
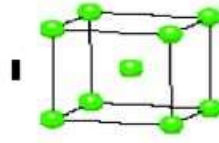
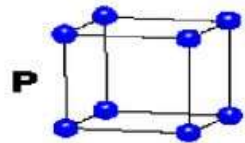


Bravais Lattice

CUBIC

$$a = b = c$$

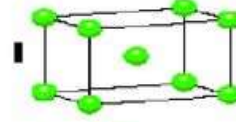
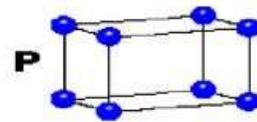
$$\alpha = \beta = \gamma = 90^\circ$$



TETRAGONAL

$$a = b \neq c$$

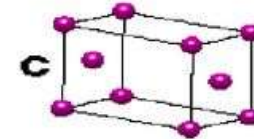
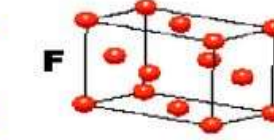
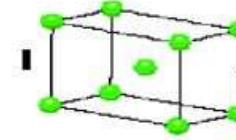
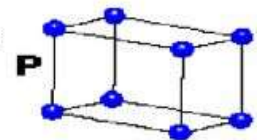
$$\alpha = \beta = \gamma = 90^\circ$$



ORTHORHOMBIC

$$a \neq b \neq c$$

$$\alpha = \beta = \gamma = 90^\circ$$

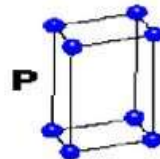


HEXAGONAL

$$a = b \neq c$$

$$\alpha = \beta = 90^\circ$$

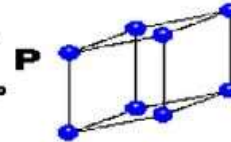
$$\gamma = 120^\circ$$



TRIGONAL

$$a = b = c$$

$$\alpha = \beta = \gamma \neq 90^\circ$$

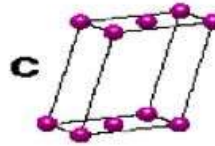
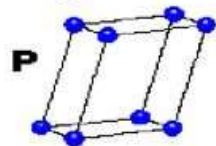


MONOCLINIC

$$a \neq b \neq c$$

$$\alpha = \gamma = 90^\circ$$

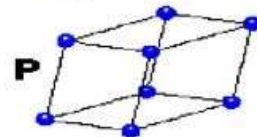
$$\beta \neq 120^\circ$$



TRICLINIC

$$a \neq b \neq c$$

$$\alpha \neq \beta \neq \gamma \neq 90^\circ$$



4 Types of Unit Cell

P = Primitive

I = Body-Centered

F = Face-Centered

C = Side-Centered

+

7 Crystal Classes

→ 14 Bravais Lattices

7 Crystal Classes with 4 possible unit cell types
Symmetry indicates that only 14 3-D lattice types occur

Bravais Space Lattices

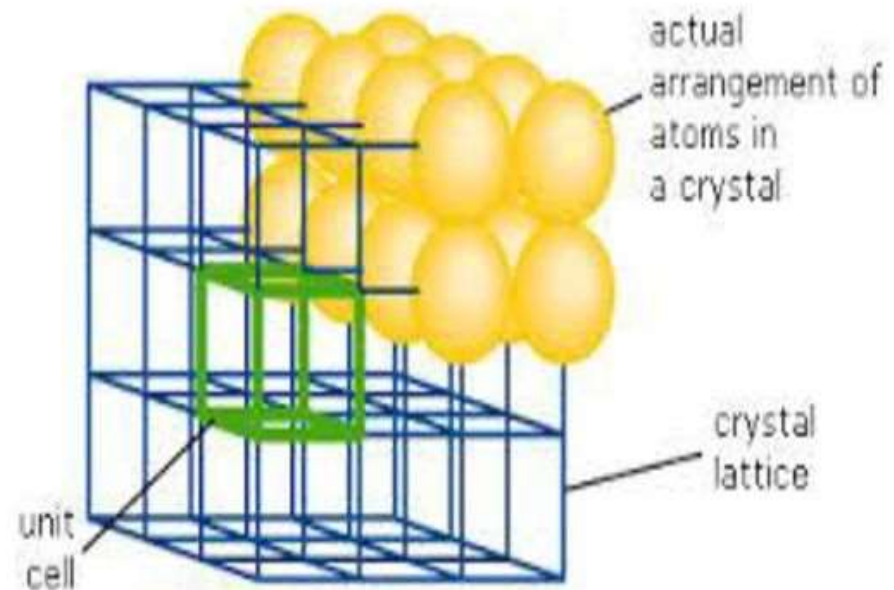
Conventionally, the finite representation of space lattices is done using **unit cells** which show maximum possible symmetries with the smallest size.

Symmetries:

1. Translation
2. Rotation
3. Reflection

Crystal Lattice.

a regular three dimensional arrangement of points in space is called a **crystal lattice**.



Bravais Lattices

There are only 14 possible three dimensional lattices. These are called **Bravais Lattices** (after the French mathematician who first described them). The following are the characteristics of a crystal lattice:

1. Primitive Lattice : P

In this lattice, the unit cell consists of eight corner atoms and all these corner atoms contribute only one effective atom for the lattice

2. Body Centred Lattice : I

In addition to the eight corner atoms, it consists of one complete atom at the centre.

3. Face Centred Lattice : F

Along with the corner atoms, each face will have one centre atom.

4. Base Centred Lattice : C

The Base and opposite face will have centre atoms along with the corner atoms.

5. Coordination Number (N):

The number of equidistant neighbors that an atom has in a crystal lattice is known as coordination number



Atomic Packing Factor

It is the ratio between the total volume occupied by the atoms or molecules in a unit cell to the volume of the unit cell.

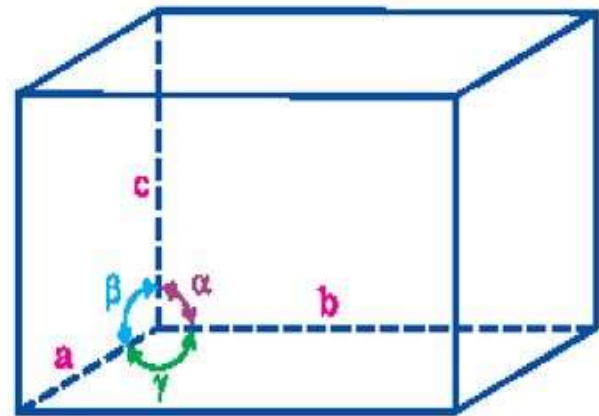
$$\text{Atomic packing factor} = \frac{\text{Total volume occupied by atoms in a unit cell}}{\text{volume of the unit cell}}$$

Atomic packing factor

$$= \frac{\text{Number of atoms present in a unit cell} \times \text{volume of the one atom}}{\text{volume of the unit cell}}$$

Unit cell

Unit cell is the smallest portion of a crystal lattice which, when repeated in different directions, generates the entire lattice.



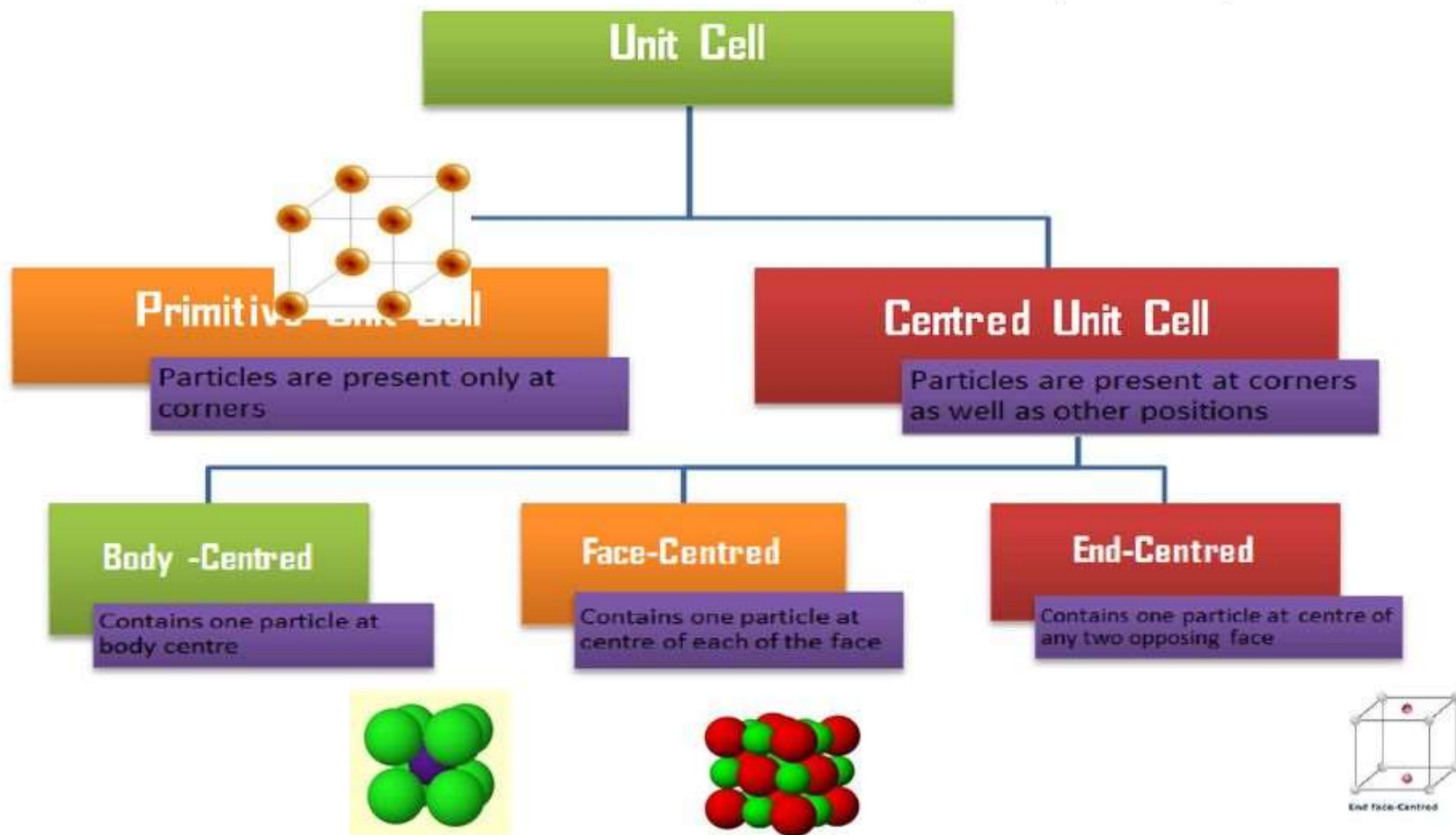
A unit cell is characterized by:

- (i) its dimensions along the three edges, a , b and c .
- (ii) angles between the edges, α (between b and c) β (between a and c) and γ (between a and b).

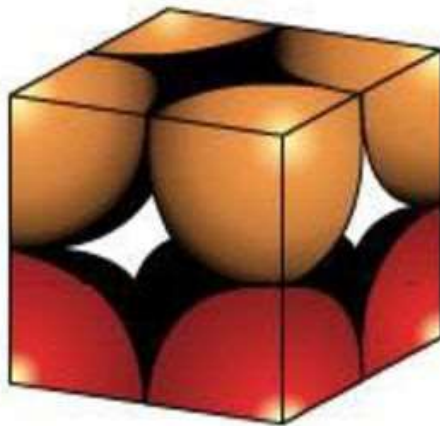
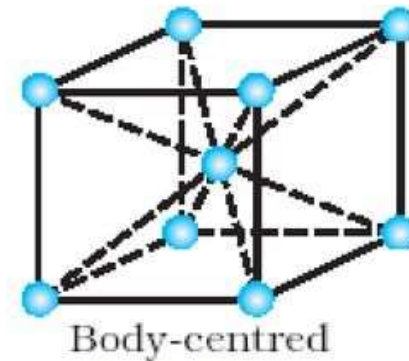
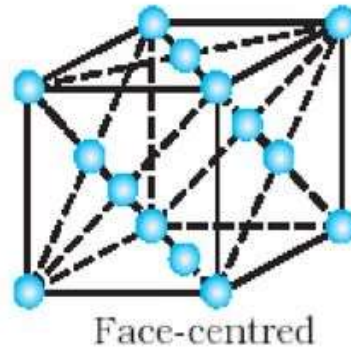
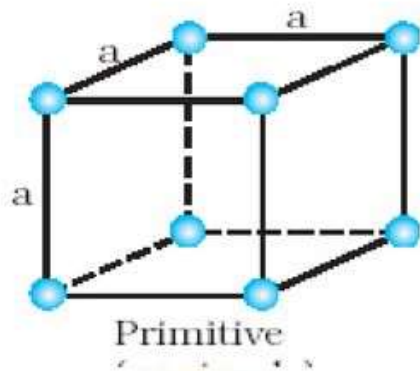
Primitive and Centred Unit Cells

(a) *Primitive Unit Cells* : the particles are present only on the corner positions of a unit cell,

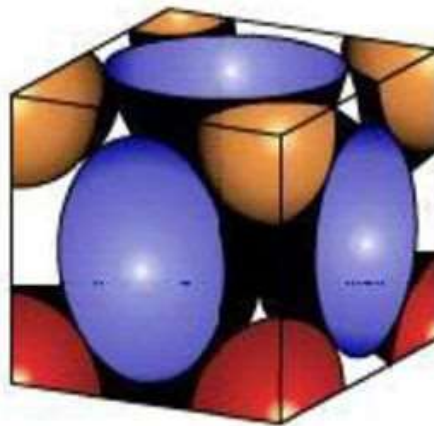
(b) *Centred Unit Cells*: contains one or more constituent particles present at positions



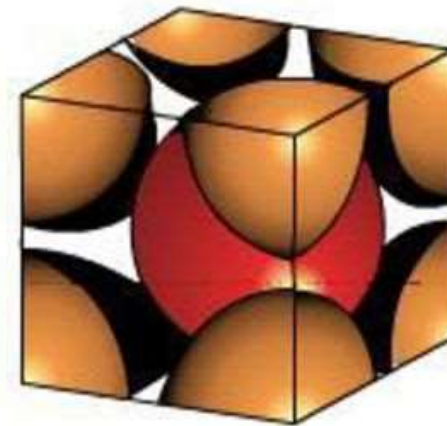
The three cubic lattices: all sides of same length, angles between faces all 90°



(a) Simple cubic



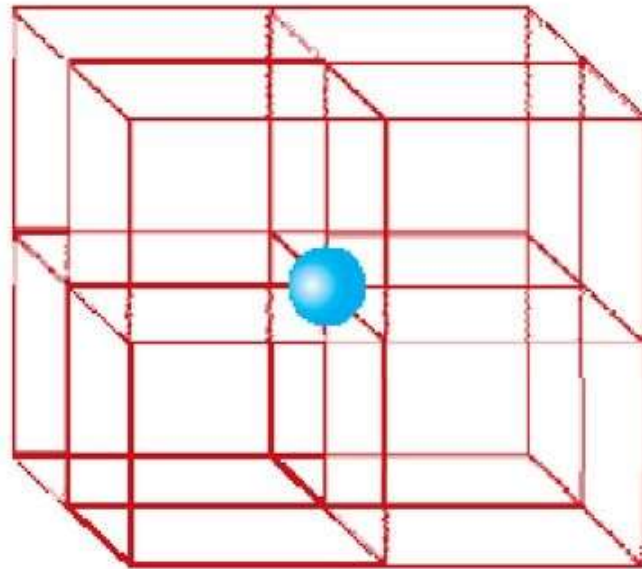
(b) Face centred cubic



(c) Body centred cubic

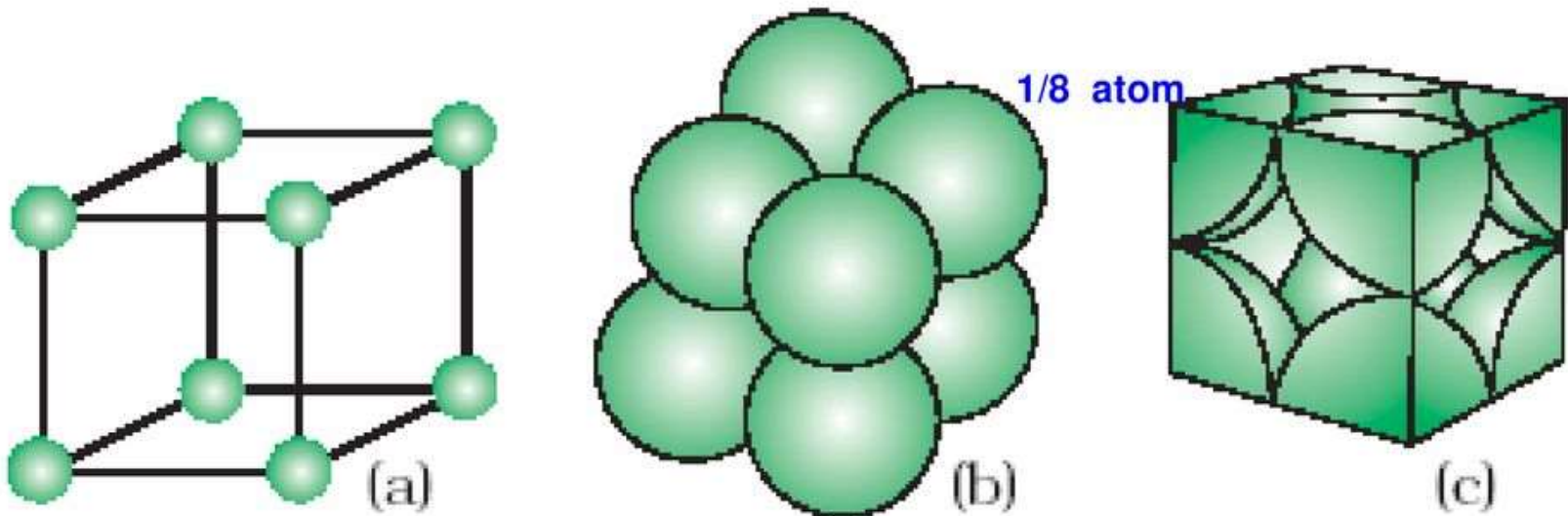
Three dimensional view showing the number of atoms per unit cell

In the following structure , each corner atom is shared between how many unit cells ?



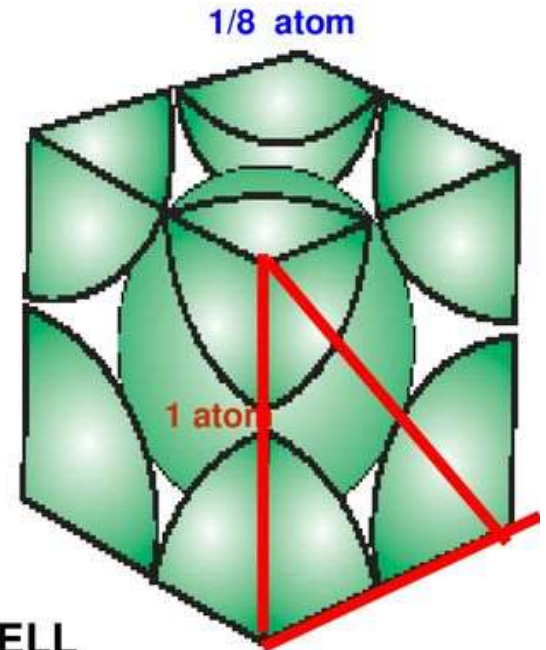
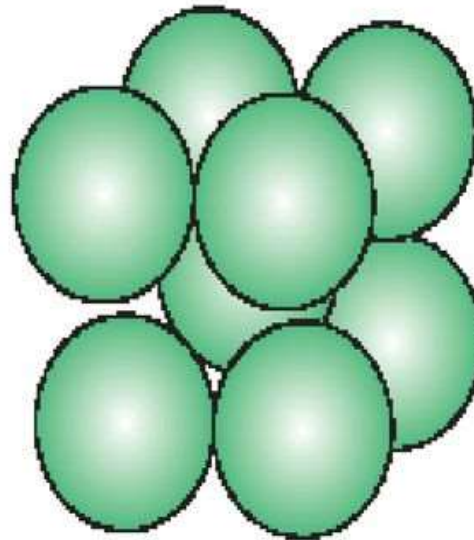
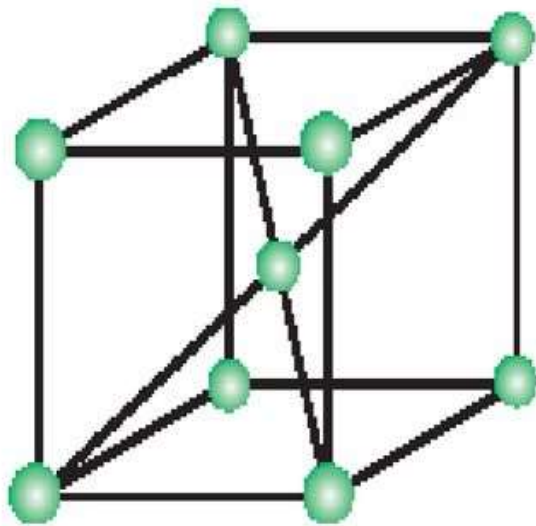
each corner atom is shared between 8 unit cells

primitive cubic unit cell (a) open structure (b) space-filling structure (c) actual portions of atoms belonging to one unit cell.



the total number of atoms in one unit cell = $8 \times \frac{1}{8} = 1$

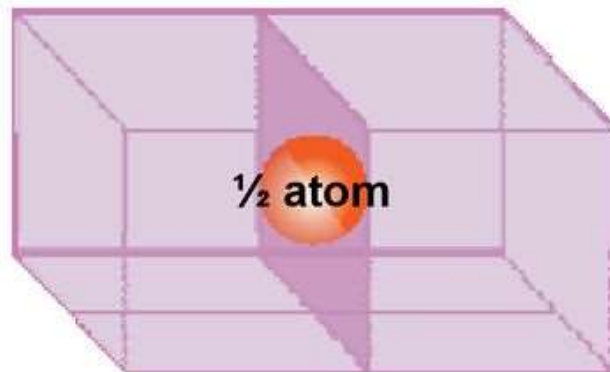
A body-centred cubic unit cell (a) open structure (b) spacefilling structure (c) actual portion of atom belonging to one unit cell



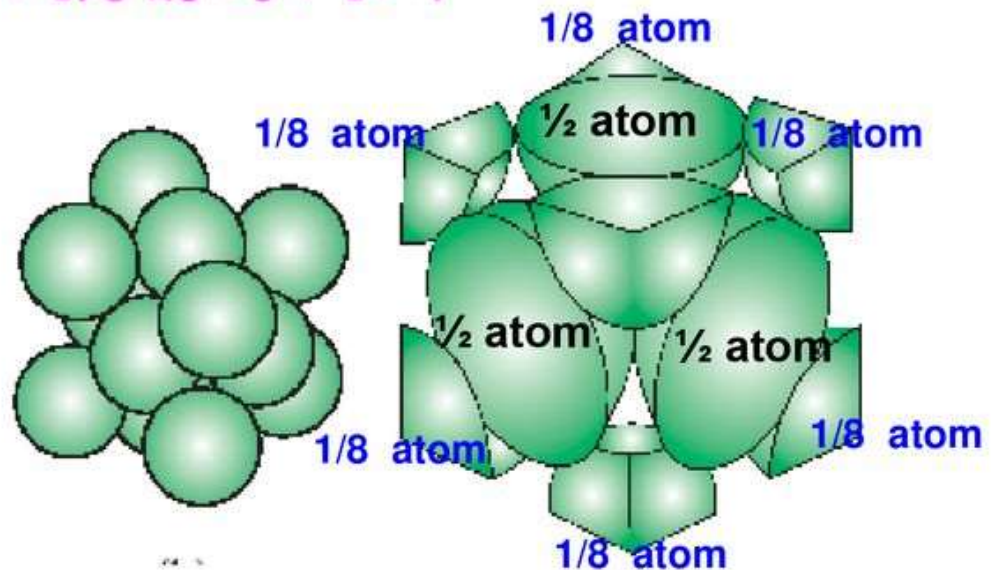
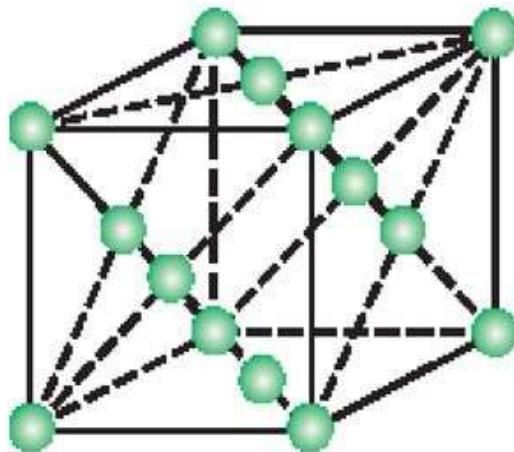
BODY CENTERED CUBIC UNIT CELL

Total no of atoms per unit cell = $1 + 1/8 \times 8 = 2$

Face- Centred Cubic Unit Cell

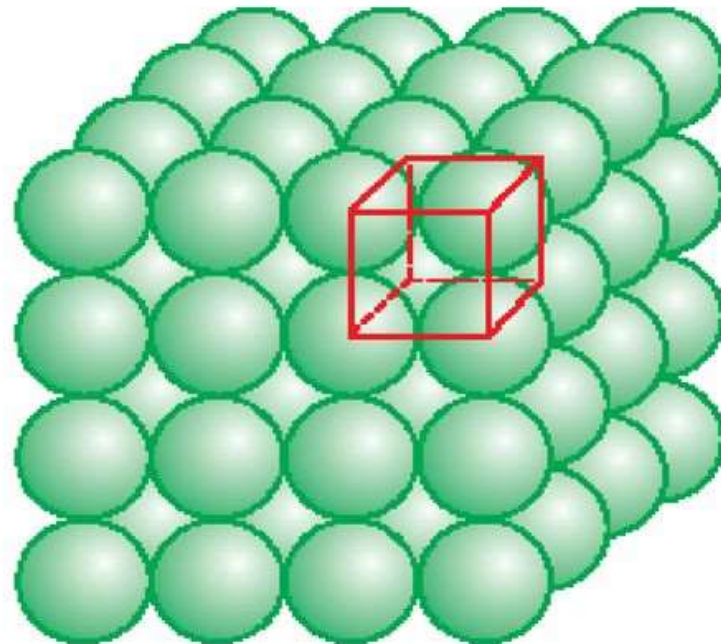


No of atoms per unit cell = $\frac{1}{2} \times 6 + \frac{1}{8} \times 8 = 3 + 1 = 4$



Close Packing in Three Dimensions

- *Three dimensional close packing from two dimensional square close-packed layers:*



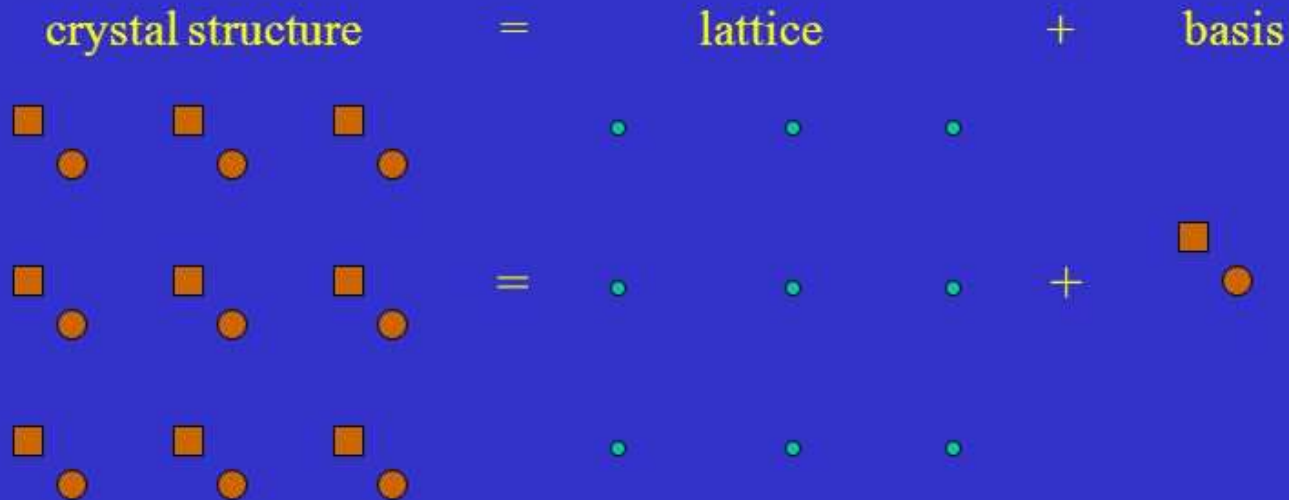
Simple cubic lattice formed by A A A arrangement

Coordination number=6

A. Lattice, Basis, and Unit Cell

An ideal crystalline solid is an infinite repetition of identical structural units in space. The repeated unit may be a single atom or a group of atoms.

An important concept:



Crystal Structure Diagrams

(a) NaCl

(b) CsCl

(c) fluorite

(d) perovskite

(e) Laves phase

(f) A15

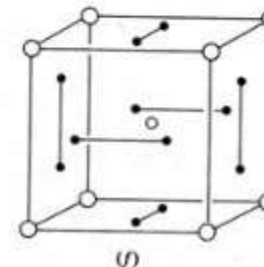
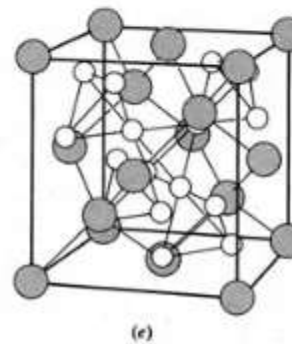
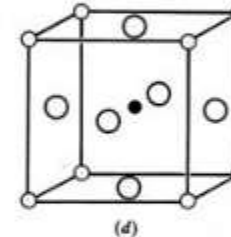
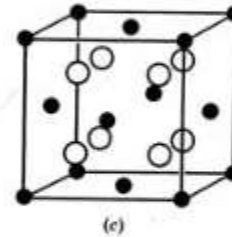
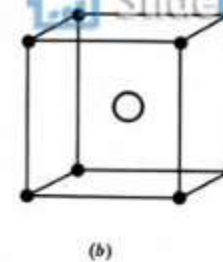
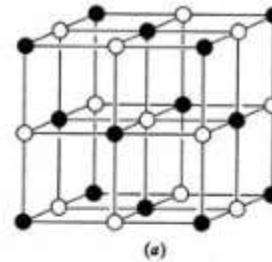
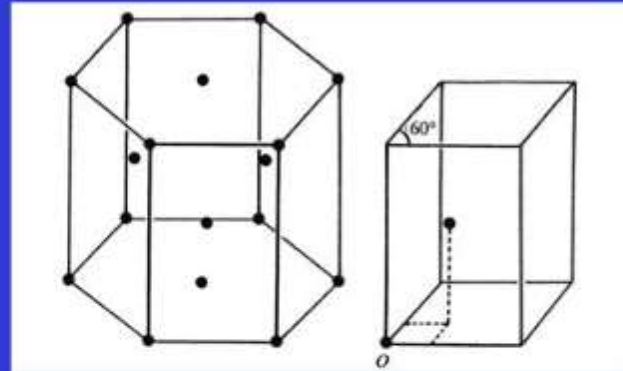


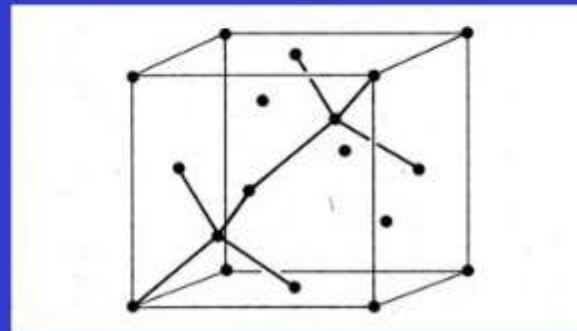
Figure 2.13 Common crystal structures: (a) NaCl; (b) CsCl; (c) fluorite (CaF_2); (d) perovskite (BaTiO_3); (e) Laves phase (e.g., Cu_2Mg); (f) A15 or β -tungsten structure (e.g., Nb_3Sn , Ta_3Sn).

Crystal Structure Diagrams (continued)

hexagonal close
packed (Be, Mg, Zn)



diamond structure
(C, Si, Ge)

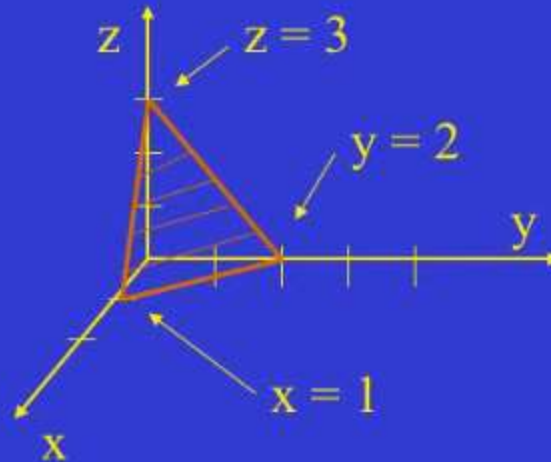


C. Miller Indices for Crystal Directions & Planes

Because crystals are usually anisotropic (their properties differ along different directions) it is useful to regard a crystalline solid as a collection of parallel planes of atoms.

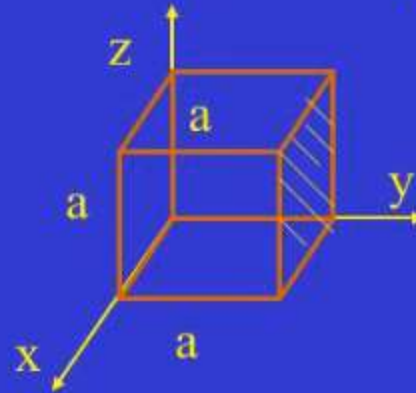
Crystallographers and CM physicists use a shorthand notation (Miller indices) to refer to such planes.

1. Determine intercepts (x , y , z) of the plane with the coordinate axes



Another example

Find the Miller indices of the shaded plane in this simple cubic lattice:



$$\bar{a} = a\hat{i} \quad \bar{b} = a\hat{j} \quad \bar{c} = a\hat{k}$$

Intercepts:

$$x = \infty \quad y = a \quad z = \infty$$

non-intersecting $\rightarrow$ intercept at ∞

Intercept ratios:

$$\frac{x}{a} = \infty \quad \frac{y}{a} = 1 \quad \frac{z}{a} = \infty$$

Reciprocals:

$$\frac{a}{x} = 0 \quad \frac{a}{y} = 1 \quad \frac{a}{z} = 0$$

We call this the (010) plane.

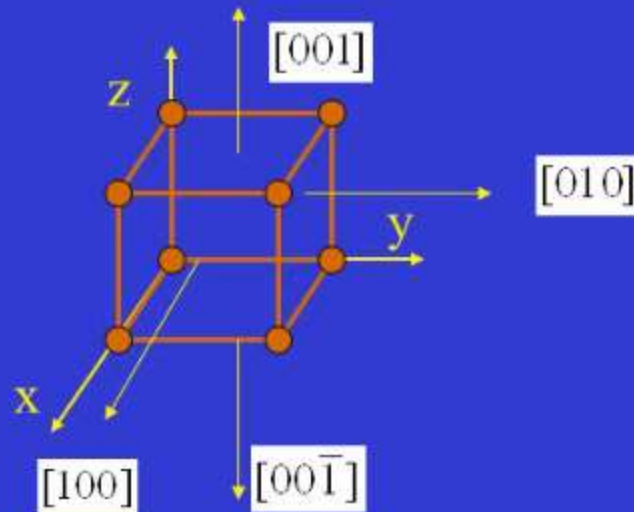
Note: (hkl) = a single plane; {hkl} = a family of symmetry-equivalent planes

Crystal Planes and Directions

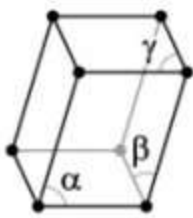
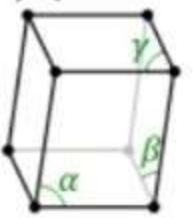
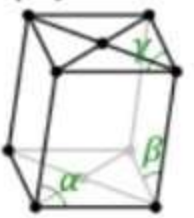
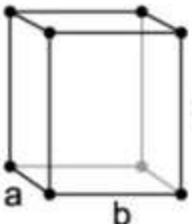
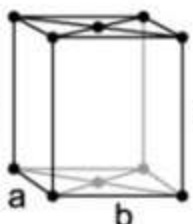
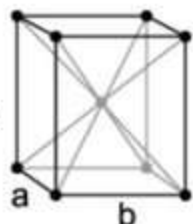
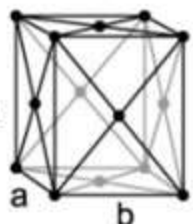
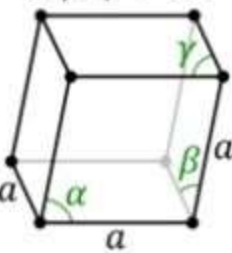
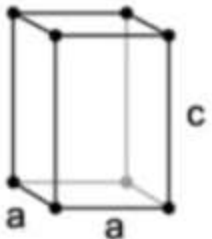
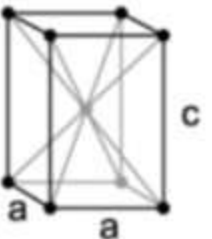
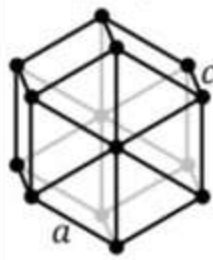
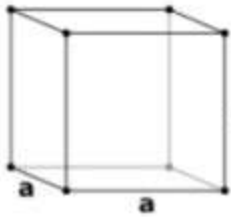
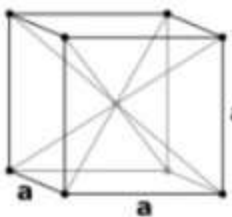
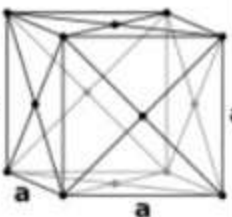


Crystal directions are specified $[hkl]$ as the coordinates of the lattice point closest to the origin along the desired direction:

Note: $[hkl]$ = a specific direction;
 $\langle hkl \rangle$ = a family of symmetry-equivalent directions



Note that for cubic lattices, the direction $[hkl]$ is perpendicular to the (hkl) plane

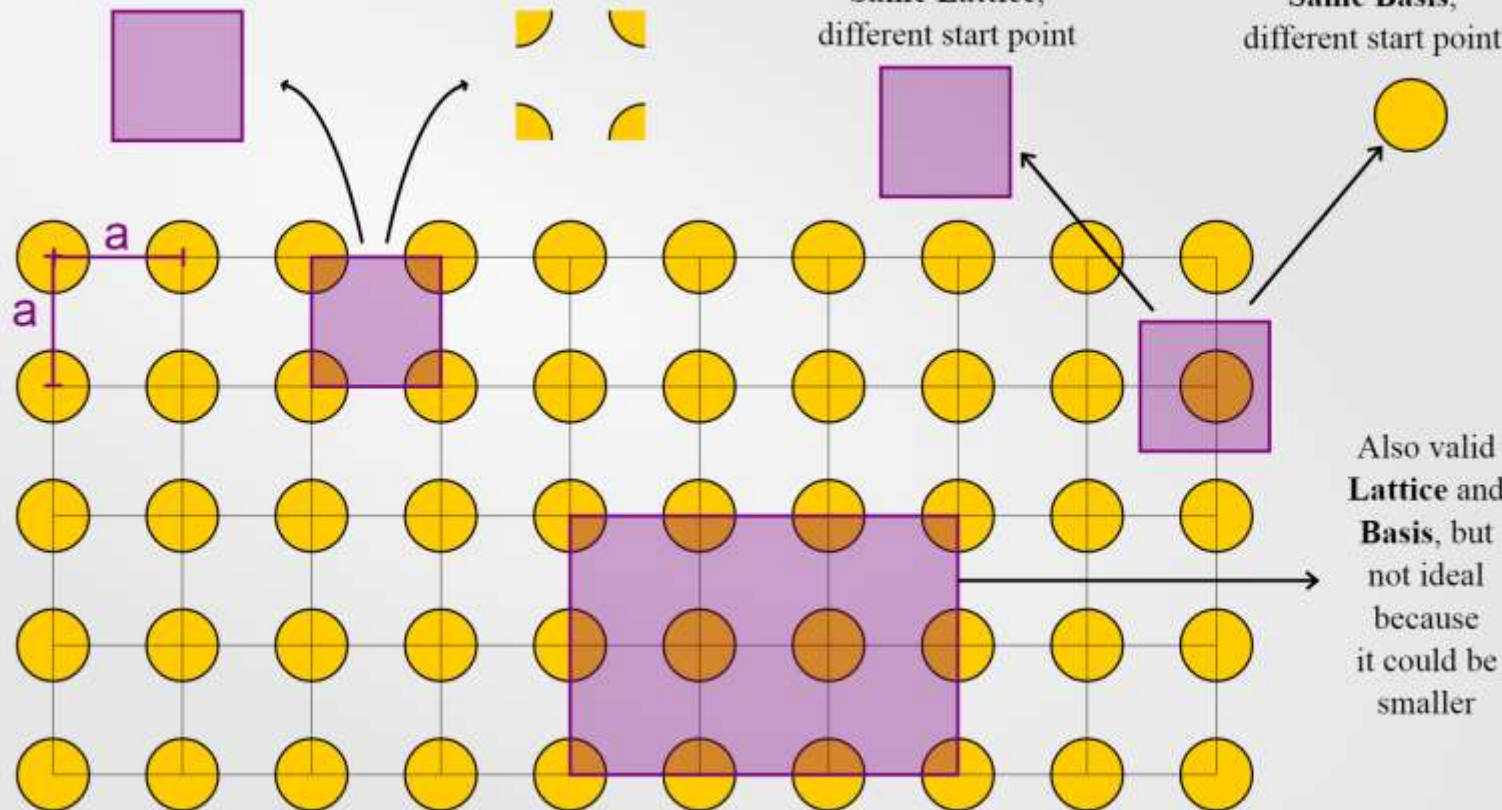
$\alpha, \beta, \gamma \neq 90^\circ$ 	$\alpha \neq 90^\circ$ $\beta, \gamma = 90^\circ$  Centered	$\alpha \neq 90^\circ$ $\beta, \gamma = 90^\circ$  Simple	$a \neq b \neq c$  Simple	$a \neq b \neq c$  Base Centered	$a \neq b \neq c$  Face Centered	$a \neq b \neq c$  Body Centered
Triclinic	Monoclinic		Orthorhombic			
$\alpha, \beta, \gamma \neq 90^\circ$ 	$a \neq c$  Simple	$a \neq c$  Body Centered	$a \neq c$ 	 Simple	 Body Centered	 Face Centered
Rhombohedral	Tetragonal		Hexagonal	Cubic (or isometric)		

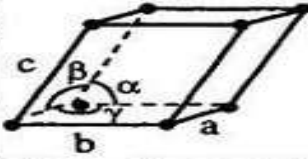
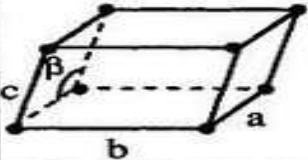
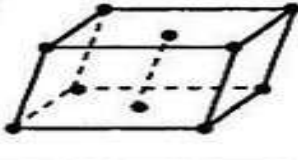
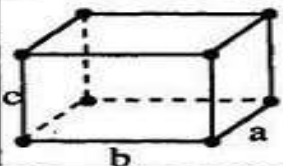
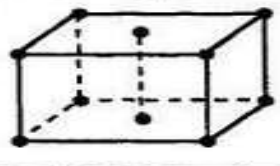
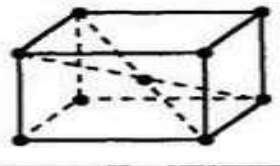
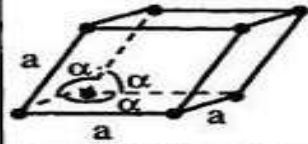
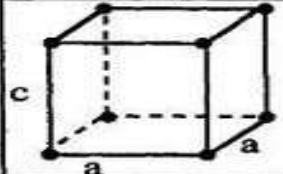
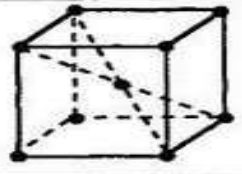
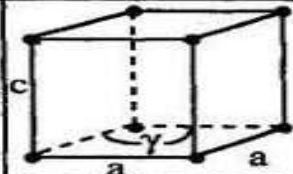
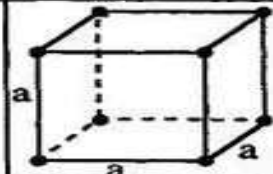
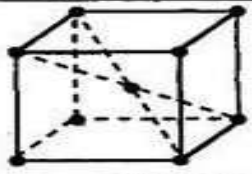
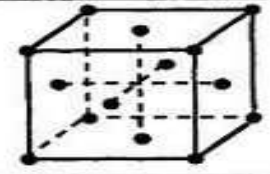
Lattice

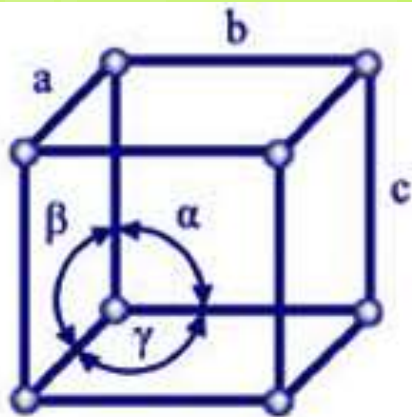
Basis

Same Lattice,
different start point

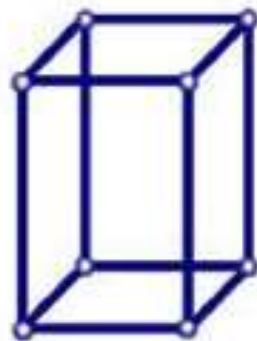
Same Basis,
different start point



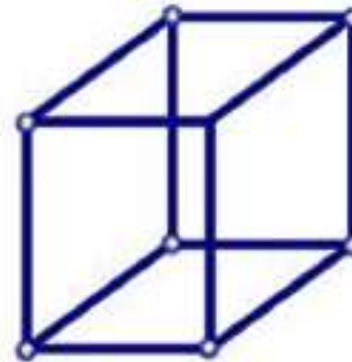
Crystal system	Bravais lattices			
	primitive	base-centered	body-centered	face-centered
Triclinic $a \neq b \neq c$ $\alpha \neq \beta \neq \gamma$				
Monoclinic $a \neq b \neq c$ $\alpha = \gamma = \frac{\pi}{2} \neq \beta$	 			
Orthorhombic $a \neq b \neq c$ $\alpha = \beta = \gamma = \frac{\pi}{2}$	   			
Trigonal $a = b = c$ $\alpha = \beta = \gamma \neq \frac{\pi}{2}$				
Tetragonal $a = b \neq c$ $\alpha = \beta = \gamma = \frac{\pi}{2}$	 			
Hexagonal $a = b \neq c$ $\alpha = \beta = \frac{\pi}{2}$ $\gamma = \frac{2\pi}{3}$				
Cubic $a = b = c$ $\alpha = \beta = \gamma = \frac{\pi}{2}$	  			



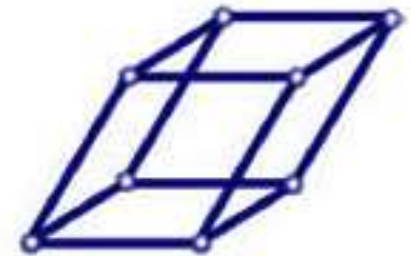
Simple Cubic
 $a = b = c$
 $\alpha = \beta = \gamma = 90^\circ$



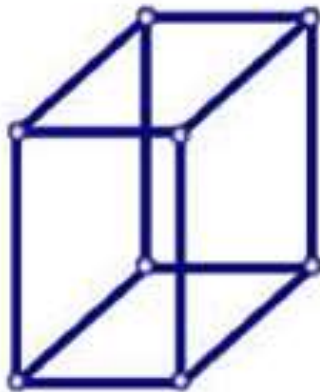
Tetragonal
 $a = b \neq c$
 $\alpha = \beta = \gamma = 90^\circ$



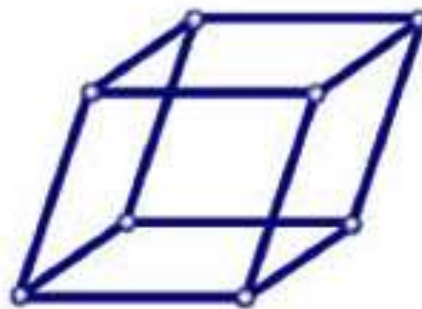
Orthorhombic
 $a \neq b \neq c$
 $\alpha = \beta = \gamma = 90^\circ$



Rhombohedral
 $a = b = c$
 $\alpha = \beta = \gamma \neq 90^\circ$



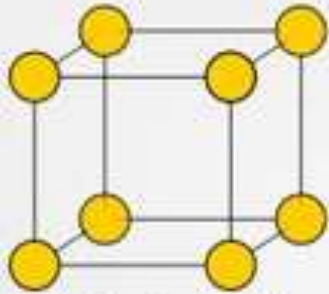
Monoclinic
 $a \neq b \neq c$
 $\gamma \neq \alpha = \beta = 90^\circ$



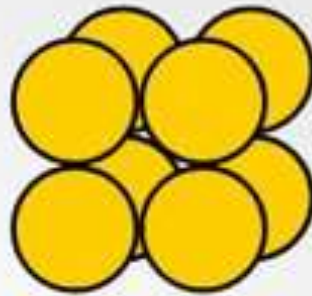
Triclinic
 $a \neq b \neq c$
 $\alpha \neq \beta \neq \gamma \neq 90^\circ$



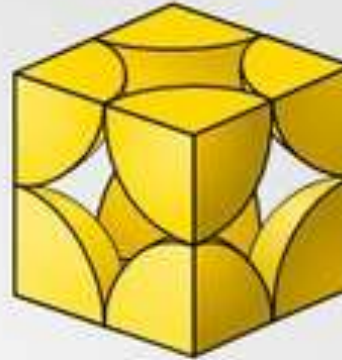
Hexagonal
 $a = b \neq c$
 $\alpha = \beta = 90^\circ, \gamma = 120^\circ$



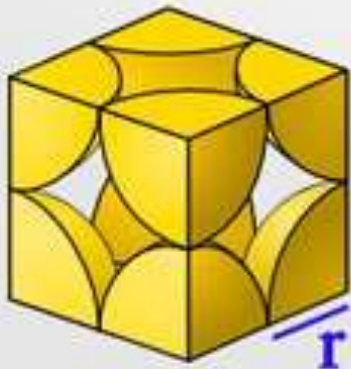
Simple Cubic
Open Visualization



Simple Cubic
Hard Sphere Model



Cut-out showing
the amount of
each atom strictly
inside the unit cell



$2r$

8 atoms on each corner,
but each atom only has
 $1/8$ of its volume inside
the unit cell

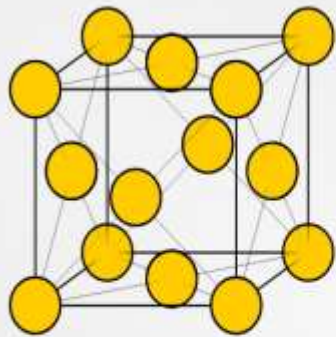
$$\text{APF} = \frac{\text{Volume of atoms}}{\text{Volume of unit cell}}$$

$$\text{Volume of atoms} = \frac{8 \cdot \frac{4}{3} \pi r^3}{8}$$

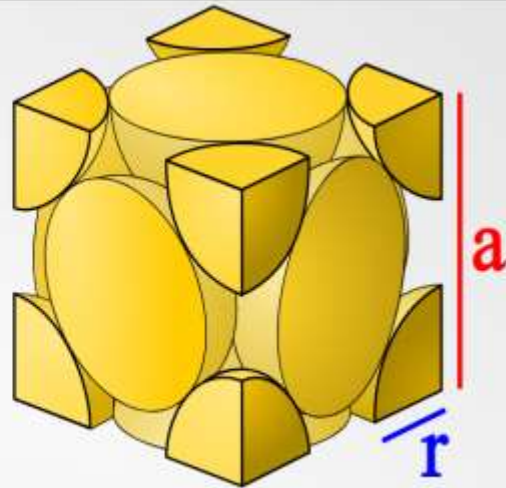
$$\text{Volume of cube} = 8r^3$$

For Simple Cubic

$$\text{APF} = \frac{\pi}{6} = 52\%$$



Face-Centered Cubic
Open Visualization



8 atoms on each corner,
with 1/8 of its volume
inside the unit cell

(1)

+ 6 atoms on each face
with 1/2 of its volume
inside the unit cell

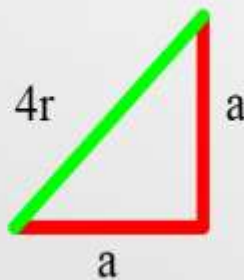
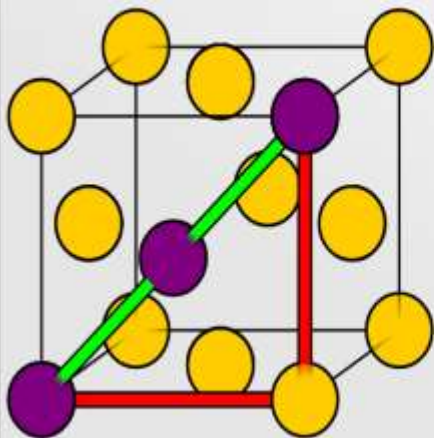
(3)

= 4 atoms per unit cell

$$APF = \frac{\text{Volume of atoms}}{\text{Volume of unit cell}}$$

Volume of atoms = $4 \cdot \frac{4}{3} \pi r^3$

Volume of cube = $(2\sqrt{2}r)^3$



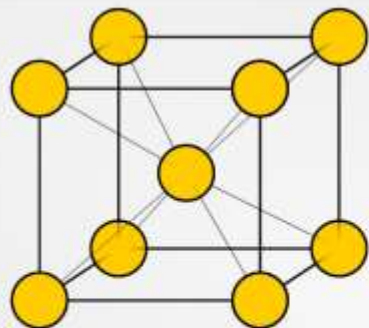
$$a^3 = 16\sqrt{2}r^3$$

$$2a^2 = (4r)^2$$

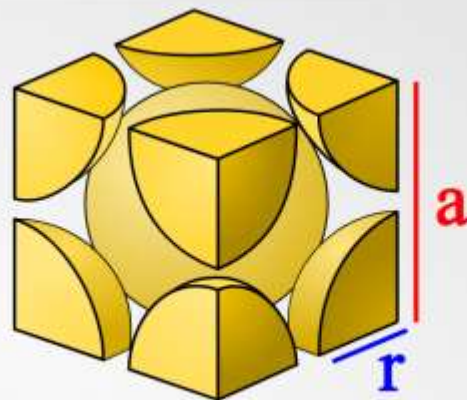
$$a = 2\sqrt{2}r$$

For FCC

$$APF = \frac{\pi}{3\sqrt{2}} = 74\%$$



Body-Centered Cubic
Open Visualization



8 atoms on each corner,
with 1/8 of its volume
inside the unit cell

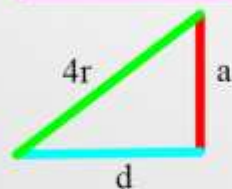
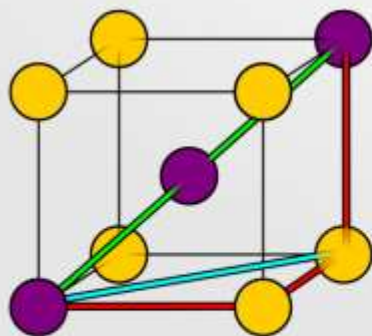
+1 full atom inside

= 2 atoms per unit cell

$$\text{APF} = \frac{\text{Volume of atoms}}{\text{Volume of unit cell}}$$

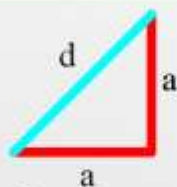
Volume of atoms = $2 \cdot \frac{4}{3} \pi r^3$

Volume of cube = $\left(\frac{4}{\sqrt{3}} r\right)^3$



$$a^2 + d^2 = (4r)^2$$

$$d^2 = 2a^2$$

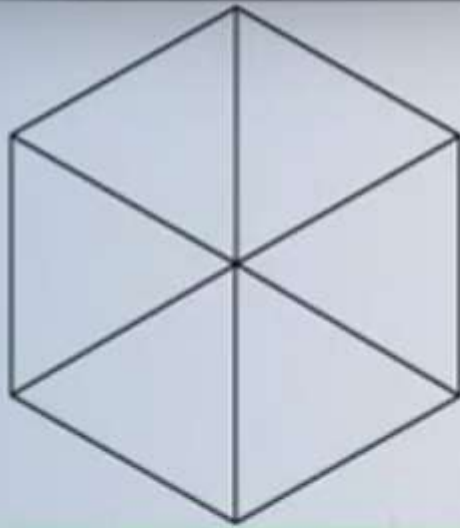


$$a^3 = \left(\frac{4}{\sqrt{3}} r\right)^3$$

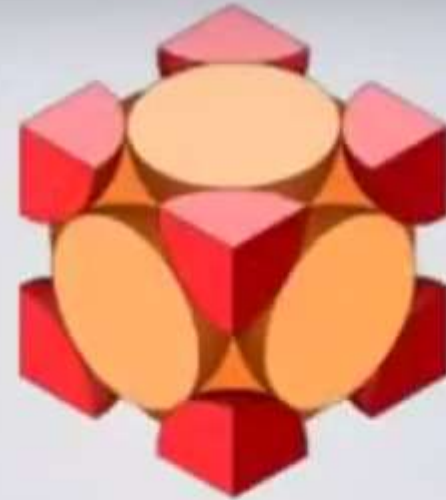
$$a = \frac{4}{\sqrt{3}} r$$

For BCC

$$\text{APF} = \frac{\sqrt{3} \pi}{8} = 68\%$$



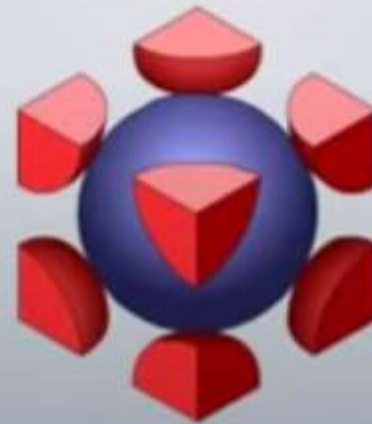
Cubic Crystal System



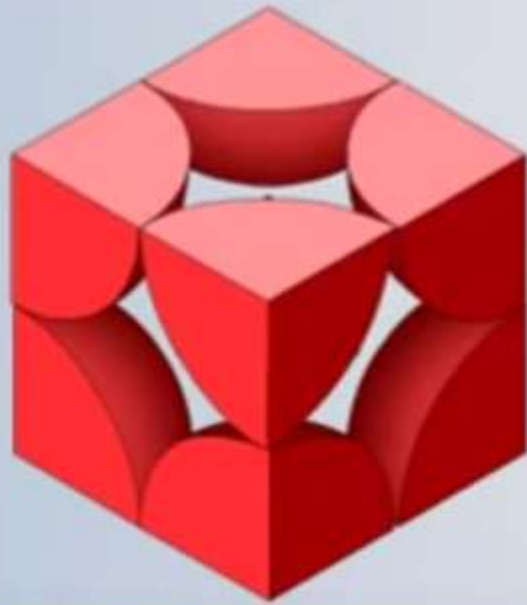
Face-Centered Cubic Lattice



Simple Cubic Lattice

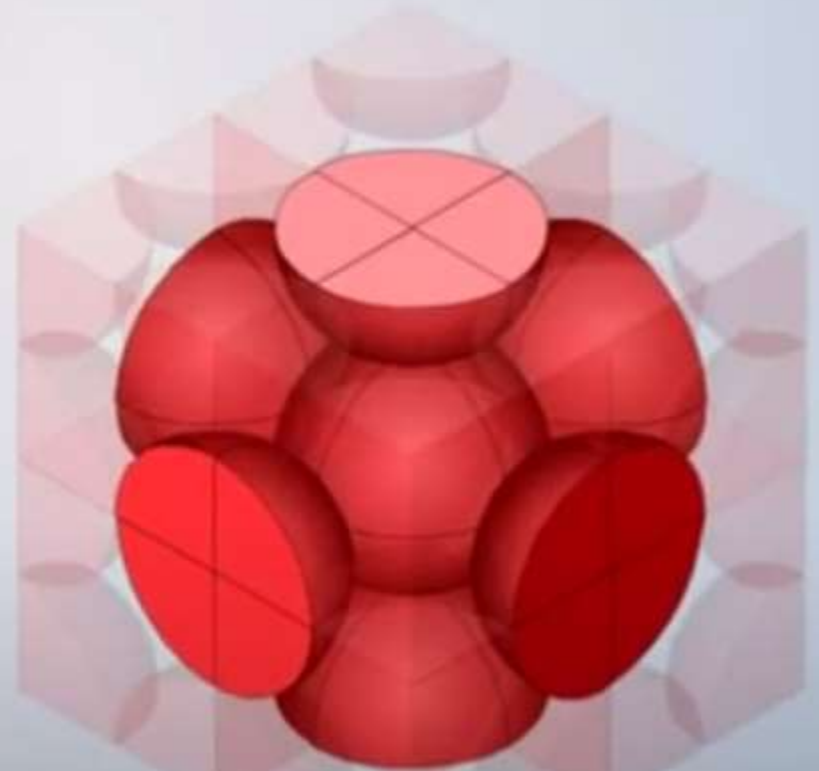


Body-Centered Cubic Lattice

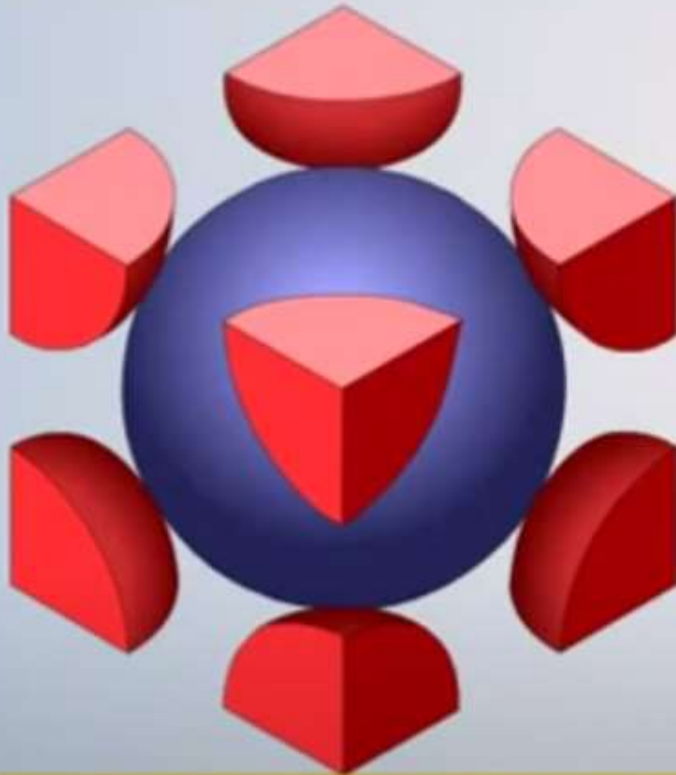


Simple Cubic Lattice

Crystal Lattice	Coordination Number
Simple Cubic (SC)	6



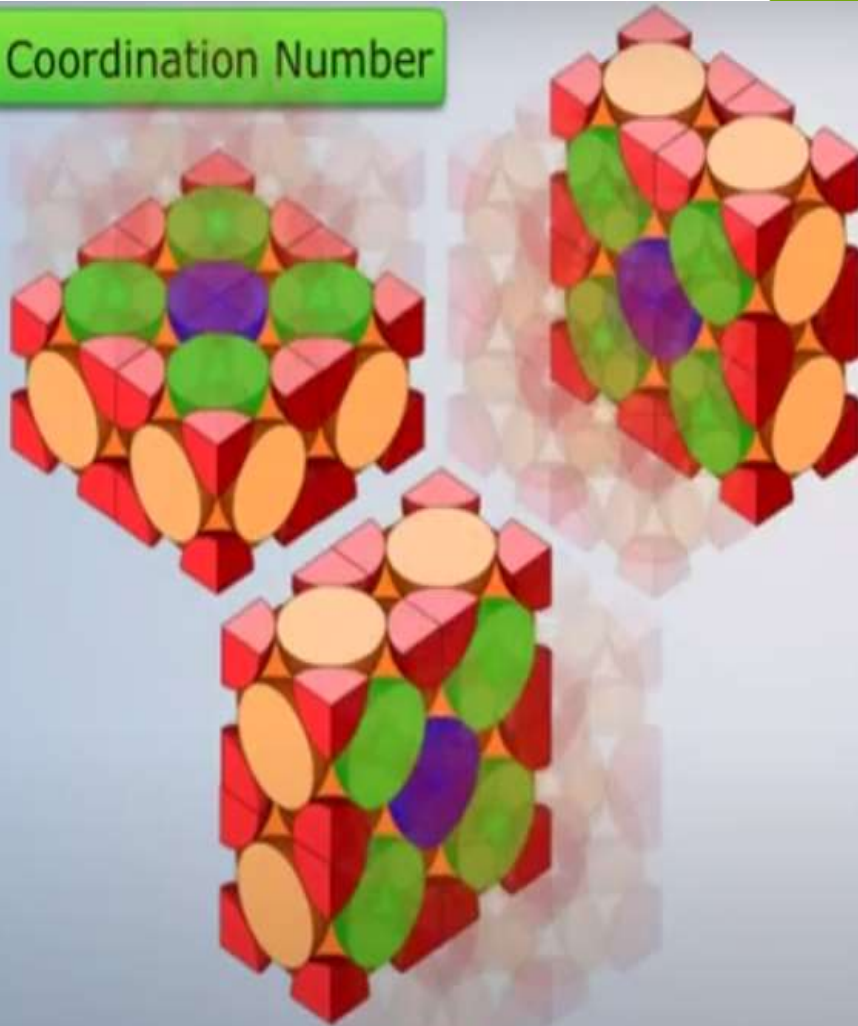
Atoms



Body-Centered Cubic Lattice

Crystal Lattice	Coordination Number
Simple Cubic (SC)	6
Body-Centered Cubic (BCC)	8

Coordination Number



► *Equi-Distant Nearest Atoms*

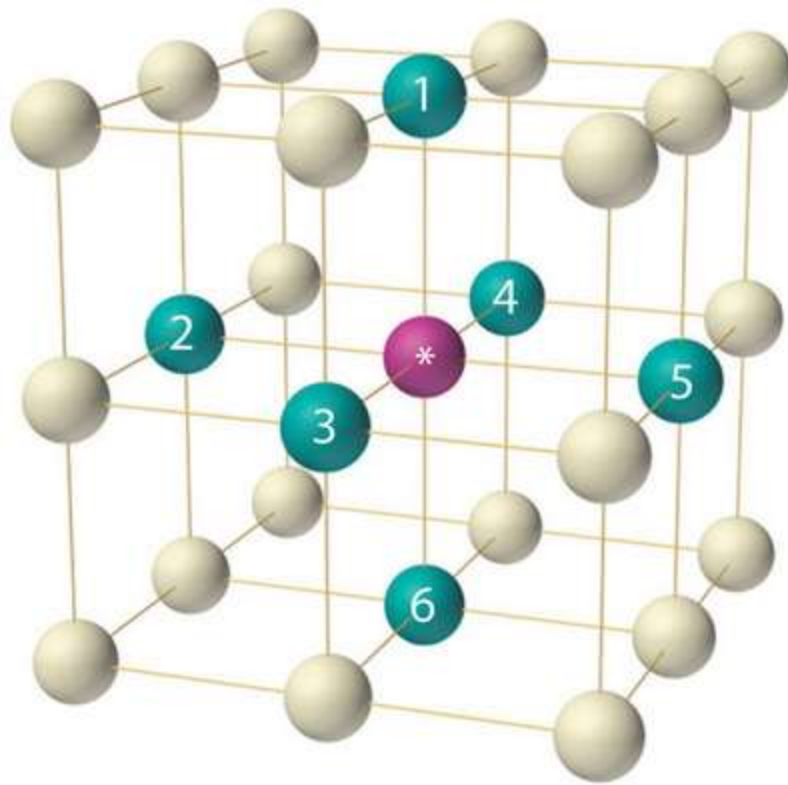
Crystal Lattice	Coordination Number
Simple Cubic (SC)	6
Body-Centered Cubic (BCC)	8
Face-Centered Cubic (FCC)	12

COORDINATION NUMBER

- ▶ *The coordination number is defined as the number of nearest neighbors surrounding the central or a given atom within the space lattice.*
- ▶ Simple cubic- coordination number is **6**
- ▶ Body centered cubic – coordination number is **8**
- ▶ Face centered cubic – coordination number is **12**

Number of atoms per unit cell

- ▶ Simple cubic - $(1/8 \times 8) = 1$
 - ▶ Body centered cubic - $(1/8 \times 8) + 1 = 2$
 - ▶ Face centered cubic - $(1/8 \times 8) + (1/2 \times 6) = 4$
- 



Body-Centered Cubic (BCC)

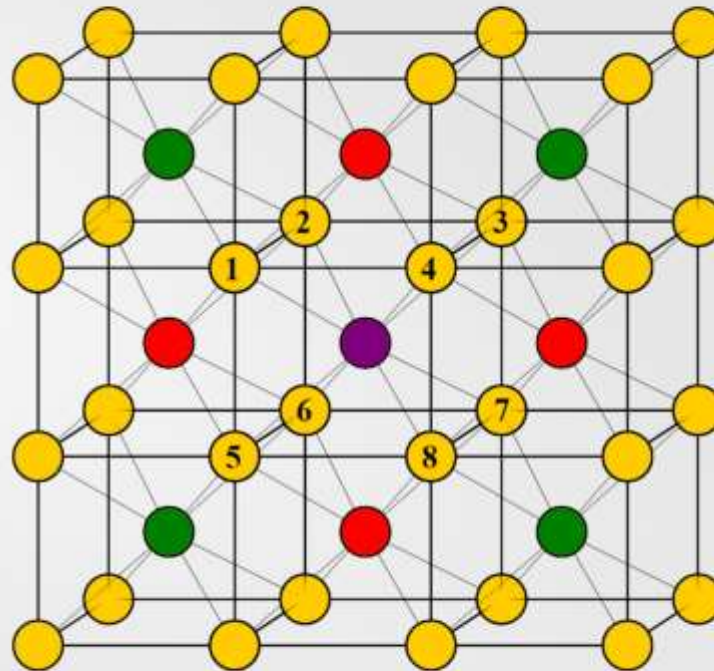
Coordination number: 8

1, 2, 3, 4, 5, 6, 7, 8

Nearest neighbors (NN)

Next-nearest
neighbors (NNN)

Next-next nearest
neighbors (NNNN)



MILLER INDICES OF THE CRYSTAL PLANES IN A UNIT CELL

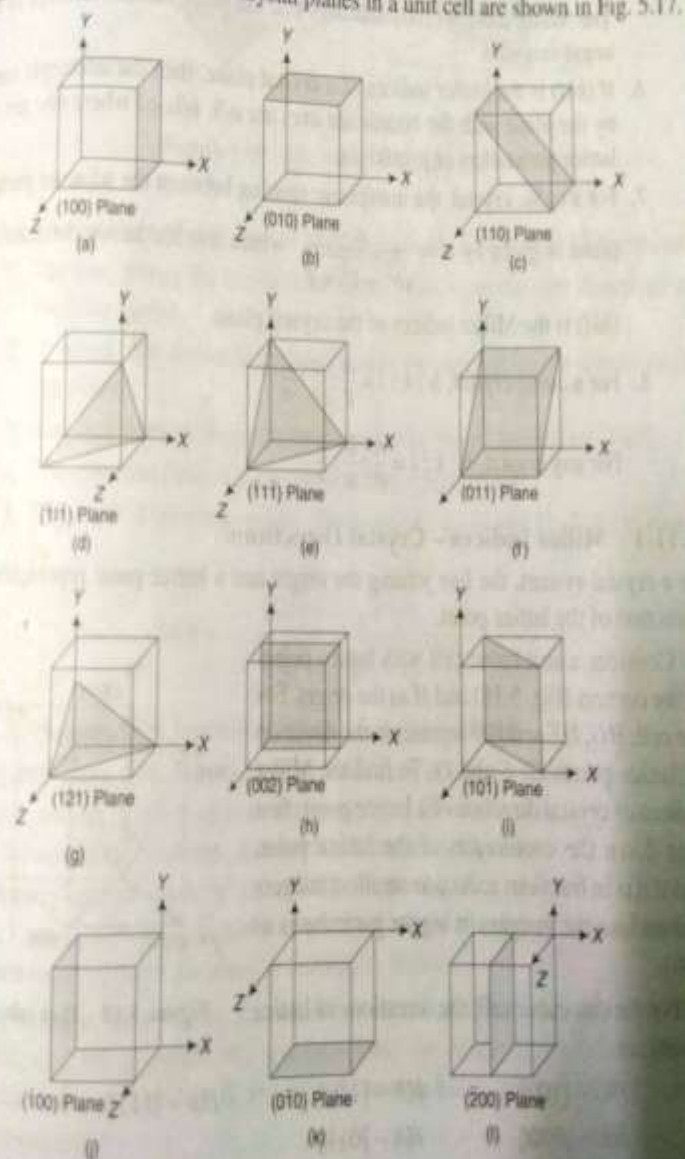
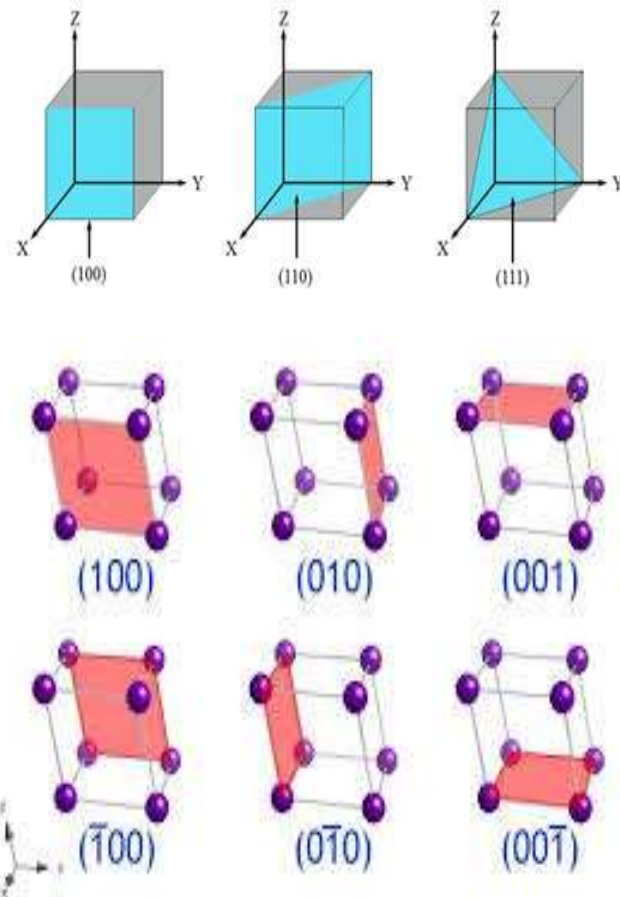


Figure 5.17 Miller indices of some crystal planes in unit cubic cell



X-ray Diffraction by Crystal planes

- ✓ For diffraction size of the obstacle must be comparable with wavelength of the incident light. Generally, diffraction grating – used for diffraction studies.
- ✓ The grating consists of a large number of closely spaced ruled lines so that the spacing
- ✓ between the lines of the order of wavelength of light.
- ✓ Due to very small wavelength (0.1 nm) of X-rays, grating cannot be used for X-ray diffraction studies as the spacing cannot be comparable to wavelength of X-rays. Laue suggested that a crystal which consists of 3d array of regularly spaced atoms will produce observable diffraction effects for X-rays. Friedrich and Knipping, succeeded in diffracting X-rays by passing X-rays through a thin crystal zinc blend.
- ✓ The diffraction pattern obtained consists of a central spot and a series of spots arranged in a definite pattern around the central spot. This pattern is known as Laue's pattern.
- ✓ The interpretation of the diffraction effect was given by Bragg.

Bragg's Law:

Consider a crystal which consists of equidistant parallel planes with the interplanar spacing d as shown figure. When X-rays of wavelength (λ) fall on these crystal planes, the atoms in the planes diffract (reflect) the X-rays in all directions,

Let the X-ray PA incident at an angle θ with the reflected at atom A in the direction AR and an QB reflected at atom B in the direction BS . These X-rays will interfere constructively or destructively upon the path difference between the X-rays.

To calculate the path difference, two normal A drawn from A to QB and BS .

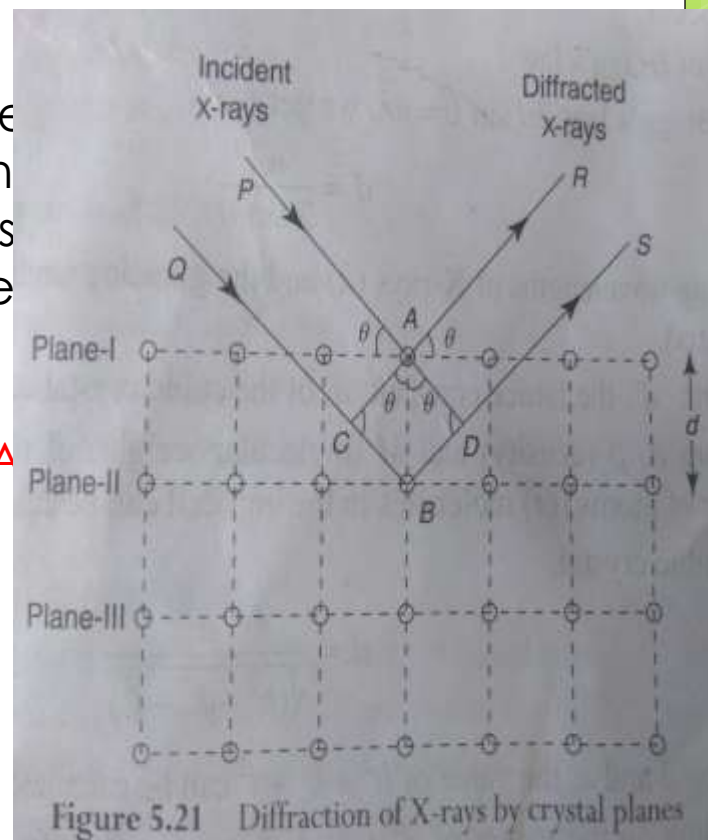
Path difference = $CB+BD$

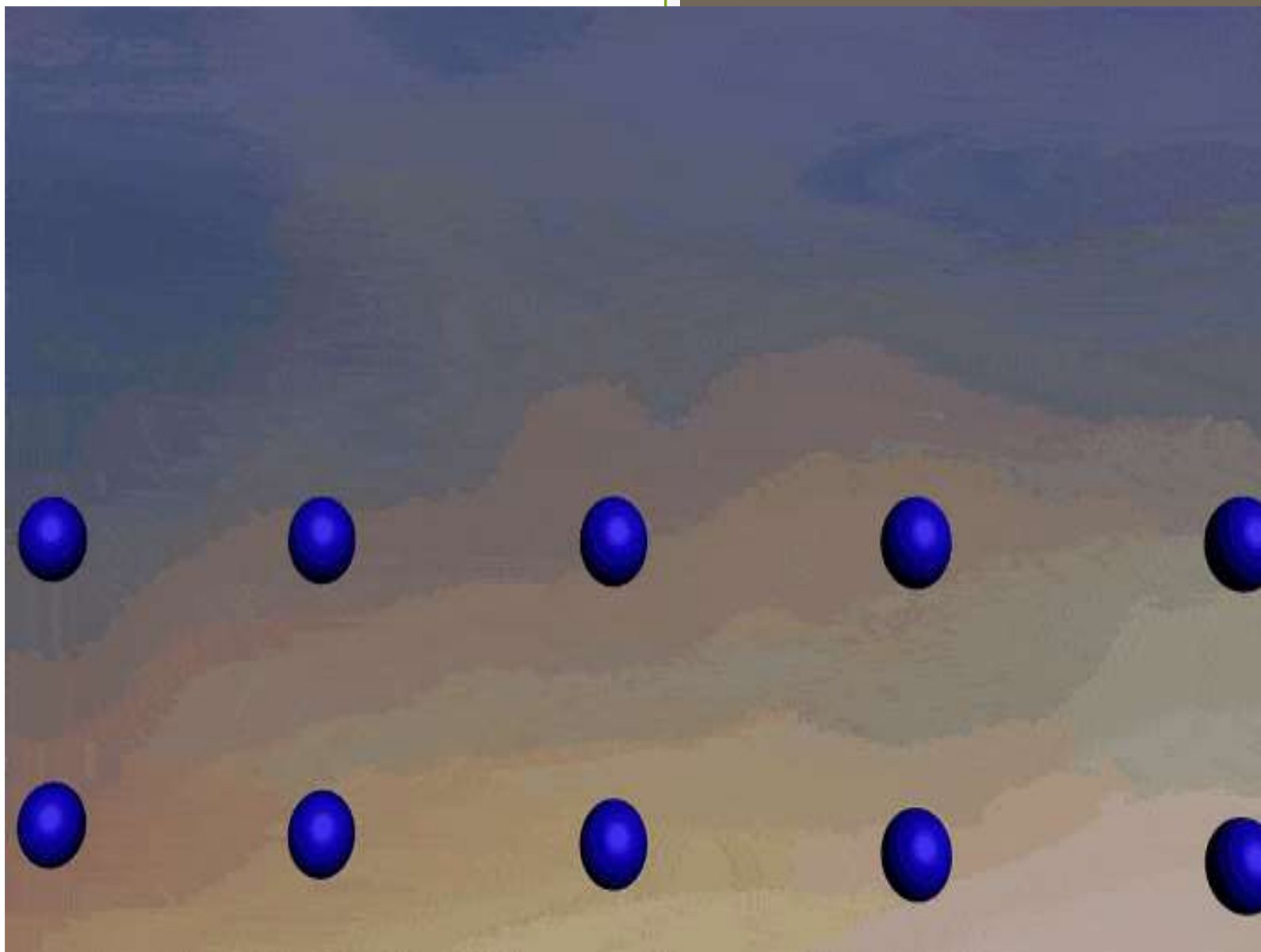
$$\text{From } \triangle ACB, \sin \theta = \frac{CB}{AB}$$

$$CB = AB \sin \theta = d \sin \theta$$

$$\text{From } \triangle ADB, BD = d \sin \theta$$

$$\text{Path difference} = d \sin \theta + d \sin \theta = 2d \sin \theta$$





To get maximum intensity, path difference should be equal to integral multiple of λ

$$\text{Path difference} = n\lambda$$

$$2d \sin \theta = n\lambda \quad \text{where } n=1,2,3$$

The above expression represents Bragg's law, according to which the diffracted X-rays interfere constructively

When the path difference is an integral multiple of the wavelength of X-rays.

Corresponding to $n=1,2,3$ we obtain first, second, third.....order diffraction spots. Since maximum possible

Value of θ is 90° , we get

$$n\lambda = 2d$$

$$\lambda \leq 2d$$

Thus, the wavelength λ should not exceed twice the interplanar spacing for diffraction to occur.

Importance of Bragg's law

1. From Bragg's law, $2d \sin \theta = n\lambda$, we get $d = \frac{n\lambda}{2 \sin \theta}$

$$\rho = \frac{nM}{a^3 N_A}$$

$$d = \frac{a}{\sqrt{h^2 + k^2 + l^2}}$$

2. Knowing d and a , $h^2+k^2+l^2$ can be calculated

3. Depending on the value of $h^2+k^2+l^2$, we can classify the crystals as BCC, FCC etc

Powder Method or Debye-Scherrer Method:

The Powder method is an XRD technique

Used to study the structure of microcrystals or tiny crystallites in the form of powder



formation of the size and orientation of the crystallites

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